

ALEKSANDR BORODIN

 June 01, 1978
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EXECUTIVE SUMMARY

Technical Leader with 20+ years of expertise in industrial water treatment and nuclear power infrastructure. Proven track record in managing complex EPC projects and licensing of VVER-1200 reactor safety-critical equipment within the regulatory framework of international standards (EU, MSZ/NBSZ, IAEA). Expert in membrane technologies (RO, UF, EDI) and asset care management for leading global customers.



EDUCATION AND QUALIFICATION

Speciality: Department of Chemistry and Applied Chemistry in Power Engineering

Graduation – July 2000

Qualification: Heat-and-power Engineer (diploma with honors)

High Education: Ivanovo State Power Engineering University (Russia)

<http://ispu.ru/sveden/faculties/TEF>



WORK EXPERIENCE

Head of the process equipment department | Licensing Division

December 2022 – Present

Atomstroyexport JSC (ASE) – Engineering division of Rosatom, Branch in Hungary

Project: Paks-2 NPP Construction (Hungary)

Key Responsibilities & Impact:

- **Regulatory Leadership:** Directing the licensing process and ensuring technical compliance for all process equipment delivered to the Paks-2 NPP with the requirements of Russian, European (EU), Hungarian nuclear regulations (MSZ/NBSZ), IAEA safety recommendations, and EPC Contract.
- **Stakeholder Management:** Leading high-level technical interactions between the General Contractor, the Hungarian Owner (Paks II), and global Suppliers/Manufacturers to secure equipment licenses.

- **Technical Oversight:** Directing the independent technical review (ITER) of licensing documentation.
- **Methodological Strategy:** Providing expert methodological support to streamline the licensing lifecycle, from conceptual design approval to final equipment delivery.

Regional Representative | Asset Care Manager

March 2008 – June 2022

GENERAL ELECTRIC / SUEZ Water Technologies & Solutions

Scope: Industrial Water & Wastewater Treatment for Tier-1 Global Customers

Key Expertise & Impact as Asset Care Manager (2020 – 2022):

- **Strategic Contract Management:** Directed administrative and commercial operations for complex service contracts, ensuring high asset reliability and customer satisfaction.
- **Business Development:** Developed comprehensive technical and commercial proposals and formulated entry strategies to secure high-value orders in the industrial sector.
- **Revenue Protection:** Successfully managed service renewals and strategy alignment to maintain long-term partnerships with key industrial clients.

Key Expertise & Impact as Regional Representative (2008 – 2020):

- **Project Lifecycle Management:** Led full-cycle project execution, including pre-commissioning, commissioning, and start-up of large-scale water treatment facilities.
- **Operational Excellence:** Managed Capital projects, BOO (Build-Own-Operate) plants, and Mobile Water Treatment units, ensuring 24/7 operational continuity in demanding environments.
- **Technical Consulting & Diagnostics:** Performed deep-dive technical surveys and advanced troubleshooting for UF, RO, and EDI systems, providing data-driven technical reports and optimization strategies.
- **Crisis Management:** Acted as a primary technical point of contact for warranty and emergency service work, minimizing downtime for mission-critical industrial assets.

Process Engineer | Lead Project Engineer | Project Manager

August 2000 – December 2007

Biotechprogress, Saint-Petersburg

Scope: Advanced Industrial Water Treatment Solutions

Key Expertise & Innovation:

- **Technology Innovation:** Led R&D initiatives for water treatment efficiency, resulting in Patent #2257265.

- **Full-Cycle Project:** Managed design and execution of industrial systems (RO, UF, Ion Exchange) from initial site surveys to final commissioning and handover.
- **Tender & Commercial Strategy:** Spearheaded technical-commercial proposals (TCP) and tender documentation, securing new industrial contracts.
- **Field Leadership:** Executed on-site inspections, pre-commissioning, and commissioning works, ensuring systems met stringent design parameters and performance guarantees.



TECHNICAL SIGNATURE:

20+ Years of Comprehensive Expertise in Advanced Water Treatment

- **Membrane Technologies:** Expert-level mastery of Reverse Osmosis (RO), Nanofiltration (NF), and Microfiltration.
- **Ultrafiltration (UF):** Extensive experience with both Pressurized and Submerged systems for municipal and industrial applications.
- **High-Purity Water:** Deep specialization in Electrodialysis (EDI), Electrodialysis, and Mixed Bed Ion Exchange filters.
- **Ion Exchange & Filtration:** Advanced knowledge of Demineralization, Softening, and Multimedia Filtration (single and multi-layer).
- **Specialized Systems:** Design and commissioning of Membrane Bioreactors (MBR) and Mobile Water Treatment Plants for rapid/emergency deployment.



PROJECT GEOGRAPHY & TIER-1 CUSTOMERS

- **Oil & Gas / Energy:** Shell, Statoil (Mongstad), TengizChevroil (TCO), Refineries (Moscow, Antipinsky, Lukoil Volgograd, Novokuibyshevsk, Bashneft, Syzran, Tuapse), Nizhnekamsknefteorgsintez, Metka Corinth, Milazzo Refinery, Paks-2 NPP.
- **Chemical & Industry:** Yara (Ferrara), Acron, Shchekinoazot, Odessa Port Plant, Acron, Permnefteorgsintez, Arcelor Mittal, Fiat, Alstom.
- **FMCG & Food Industry:** PepsiCo, Cargill, Savushkin Product, Sodruzhestvo-Soya.
- **Infrastructure:** Elecnor (Spain), Municipal Water Authority of Dubrovnik (Croatia).



SKILLS AND PERSONAL QUALITIES

- **Technical Expertise:** Reverse Osmosis (RO), Ultrafiltration (UF), Nanofiltration (NF), Electrodialysis (EDI), Ion Exchange, Multimedia Filtration, Desalination, Mobile Water Treatment Plants, Municipal Potable Water Systems, Industrial Wastewater Treatment, Nuclear Safety Standards (IAEA/EN/MSZ/ NBSZ).

- **Project & Regulatory Management:** EPC/EPCM Project Execution, Stakeholder Management, Regulatory Compliance, Asset Care Management, Commissioning & Start-up, Troubleshooting, Service Contract Administration.
- **Software & Analytics:** SAP, Water Treatment Data Analysis & Projection Software, Strategic Planning, Decision Making, Technical Reporting.
- **Languages:** Russian - Native, English - Business Proficient, Hungarian - Beginner.
- **Soft Skills & Attributes:** Proven ability to work in ever-changing regulatory environments, Strategic thinking, Team coordination, Focus on project results, Strong diagnostic skills, Innovation.



PROFESSIONAL RECOGNITION & INNOVATION

- **Patent for Invention #2257265:** Author and owner of an innovative method for industrial water treatment, showcasing a commitment to R&D and process optimization.
- **Technical Publications:** Author of specialized articles in peer-reviewed engineering journals, focusing on membrane technologies and water chemistry in power engineering.
- **Innovative Vision:** Dedicated to implementing state-of-the-art, high-efficiency water treatment solutions that set new benchmarks for safety and sustainability in the nuclear and industrial sectors.



KEY PROJECT HIGHLIGHTS

- **Paks-2 NPP (Hungary):** Leading the complex licensing and technical compliance process for VVER-1200 reactor equipment under EU and HAEA (Hungarian) regulations.
- **Ombra Water Treatment (Croatia):** Successfully managed the commissioning and fine-tuning of advanced pressure ultrafiltration (UF) units for the city of Dubrovnik.
- **Arctic Desalination (Pevek):** Directed the emergency deployment and 24/7 operation of a mobile desalination plant under extreme sub-zero conditions for the Chaunskaya CHP plant. This critical mission ensured Pevek town reliability during the transition period prior to the commissioning of the Floating Nuclear Power Plant (FNPP).